中山大学计算机学院

人工智能实验报告

课程名称：Artificial Intelligence

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一，实验题目

运用pytorch完成药品分类

要求：

- 运用pytorch框架完成中药图片分类，具体见给出的训练集和测试集(data.zip)
- 需要画出loss、准确率曲线图。

二，实验内容

算法原理

调用`torch`来搭建神经网络，实现对药图片的分类

> 核心问题

- 数据导入
- 网络（模型）搭建
- 模型训练
- 输出结果（结果可视化）

伪代码

（主要是训练部分）

```
for epoch in total_times:
    for pic, label in train_data:
        input_data = convert(pic)
        pred = model(input_data)
        loss = judge(pred, label)

        backward()
        parameter.update()

    correct = 0
```

```
    for pic, label in test_data:
        input_data = convert(pic)
        pred = model(data)
        label_pred = argmax(pred)
        if label_pred == label:
            correct += 1
    print(correct / total)
```

关键代码展示

> 自定义数据集

```
# 输入为: data/train, data/test
class MyDataset(Dataset):
    def __init__(self, root, resize):
        super(MyDataset, self).__init__()
        self.images_path, self.labels = [], []
        self.root = root
        self.resize = resize

        # 获取到每一个图片的路径
        for i, name in enumerate(sorted(os.listdir(os.path.join(root)))):
            cur_dir = os.path.join(root, name)
            if not os.path.isdir(cur_dir):
                continue

            temp = [os.path.join(root, name, imgdir) for imgdir in
os.listdir(cur_dir)]
            self.images_path.extend(temp)
            self.labels.extend([i] * len(os.listdir(cur_dir)))
        assert len(self.images_path) == len(self.labels), 'unknown error'

    def __len__(self):
        return len(self.images_path)

    def __getitem__(self, idx):
        img, label = self.images_path[idx], self.labels[idx]

        tf = transforms.Compose([
            lambda x: Image.open(x).convert('RGB'),  # 根据路径获得彩图
            transforms.Resize((self.resize, self.resize)),  # 进行一些图像处理
            transforms.RandomRotation(0),
            transforms.CenterCrop(self.resize),
            transforms.ToTensor(),  # 将图片转为张量
            transforms.Normalize(mean=[0.485, 0.456, 0.406],  # 对应RGB通道
                                 std=[0.229, 0.224, 0.225])
        ])

        # 将图片，标签转为张量
        img = tf(img)
```

```
        label = torch.tensor(label)
        return img, label
```

> 两种神经网络结构

实践发现`stride`取1的时候显存会爆，所以只能取2

```
# 自定义卷积神经网络层
class MyCNN(nn.Module):
    def __init__(self):
        super(MyCNN, self).__init__()
        self.stride = 2
        self.conv_unit = nn.Sequential(
            nn.Conv2d(3, 6, kernel_size=(5, 5), stride=(self.stride,
self.stride)),
            nn.MaxPool2d(2, 2),
            nn.Conv2d(6, 16, kernel_size=(5, 5), stride=(self.stride,
self.stride))
        )
        tmp = torch.randn(batch_size, 3, 224, 224)
        out = self.conv_unit(tmp)
        # out [batch_size, 16, 106/26, 106/26]
        print("out:", out.shape)
        assert True, "break"
        self.fc_input_dim = 16*106*106 if self.stride == 1 else 16*26*26
        self.fc_unit = nn.Sequential(
            nn.Linear(self.fc_input_dim, 1024),
            nn.ReLU(),
            nn.Linear(1024, 128),
            nn.ReLU(),
            nn.Linear(128, 5)
        )

    def forward(self, x):
        x = self.conv_unit(x)
        length = x.size(0)
        x = x.view(length, self.fc_input_dim)
        logits = self.fc_unit(x)
        return logits


# 搭建resnet18神经网络
class Resnet18(nn.Module):
    def __init__(self, classes=5):
        super(Resnet18, self).__init__()
        self.features = models.resnet18(pretrained=True)
        self.features.fc = nn.Linear(512, classes)

    def forward(self, x):
        x = self.features(x)
        return x
```

> 训练模型

```
def main():
    # [batch, 3, len, wid]
    train_db = MyDataset('data\\train', 224)
    test_db = MyDataset('data\\test', 224)

    # assert False, 'break'
    train_loader = DataLoader(train_db, batch_size=batch_size, shuffle=True)
    test_loader = DataLoader(test_db, batch_size=batch_size, shuffle=False)
    ''' 测试输入输出维度的代码
    net = MyNet()
    tmp = torch.randn(32, 3, 224, 224)
    out = net(tmp)
    print('out:', out.shape)
    print('--------------------------------')
    '''

    # 令程序跑在GPU上，参数初始化
    assert torch.cuda.is_available(), 'cuda error'
    device = torch.device('cuda:0')
    # 选择模型并初始化
    if model_type:
        model = Resnet18().to(device)
    else:
        model = MyCNN().to(device)
    # 选择优化器并初始化
    criterion = nn.CrossEntropyLoss()
    if optim_type:
        optimizer = optim.Adam(model.parameters(), lr=learning_rate)
    else:
        optimizer = optim.SGD(model.parameters(), lr=learning_rate)
    # print(model)  # 打印模型参数

    # 开始训练
    for epoch in range(epoch_nums):
        train_average_loss, test_average_loss = 0.0, 0.0
        train_total_correct, test_total_correct = 0.0, 0.0
        train_total_num, test_total_num = 0.0, 0.0

        # train
        model.train()
        for batchidx, (x, label) in enumerate(train_loader):
            x, label = x.to(device), label.to(device)

            logits = model(x)
            pred = logits.argmax(dim=1)
            train_total_correct += torch.eq(pred, label).float().sum().item()
            train_total_num += x.size(0)
            # logits: [batch_size, 5], label: [batch_size]
            loss = criterion(logits, label)
            train_average_loss += loss.item()
```

```
            # 反向传播部分
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

        train_acc = train_total_correct / train_total_num
        train_average_loss /= len(train_loader)
        print('epoch', epoch, ': train_loss ->', train_average_loss, end='\t\t')
        print('epoch', epoch, ': train_accuracy ->', train_acc)
        loss_train_xs.append(epoch), loss_train_ys.append(train_average_loss)
        acc_train_xs.append(epoch), acc_train_ys.append(train_acc)


        # test 无需 backprop
        model.eval()
        with torch.no_grad():
            for x, label in test_loader:
                x, label = x.to(device), label.to(device)
                logits = model(x)
                pred = logits.argmax(dim=1)
                test_total_correct += torch.eq(pred, label).float().sum().item()
                test_total_num += x.size(0)

                loss = criterion(logits, label)
                test_average_loss += loss.item()

            test_acc = test_total_correct / test_total_num
            test_average_loss /= len(test_loader)
            print('epoch', epoch, ': test_loss ->', test_average_loss, end='\t\t')
            print('epoch', epoch, ': test_accuracy ->', test_acc)
            loss_test_xs.append(epoch), loss_test_ys.append(test_average_loss)
            acc_test_xs.append(epoch), acc_test_ys.append(test_acc)
            print('------------------------------------------------------------')
```

> 可调整参数

```
# 模型可调式参数
batch_size = 16
learning_rate = 0.001
epoch_nums = 20
model_type = 1  # 卷积网络或者resnet18
optim_type = 0  # Adam或者SCG优化器
```

创新点 & 优化

1. 统一训练集测试集数据结构，方便统一读取
2. 设计两个网络模型以及多个可选参数，便于比较性能
3. 采取GPU运行，加速运行过程（本来没安装cuda）

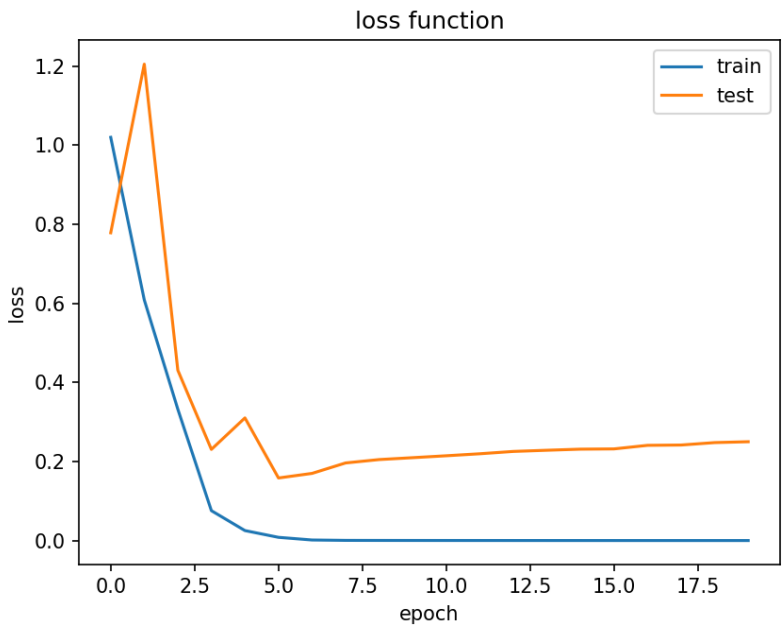
三，实验结果及分析

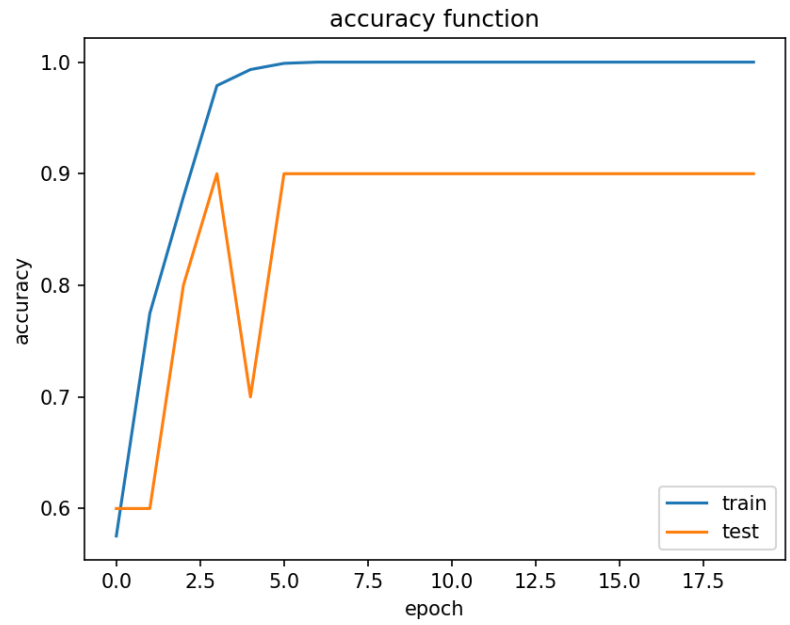
根据测试，发现`learning rate`取0.001，`batch_size`设为16时的效果较好。同时在设定的两个模型以及两种优化器中，卷积网络搭配`Adam`，`Resnet18`搭配`SGD`可以取得很好的效果。

> 卷积网络 + Adam

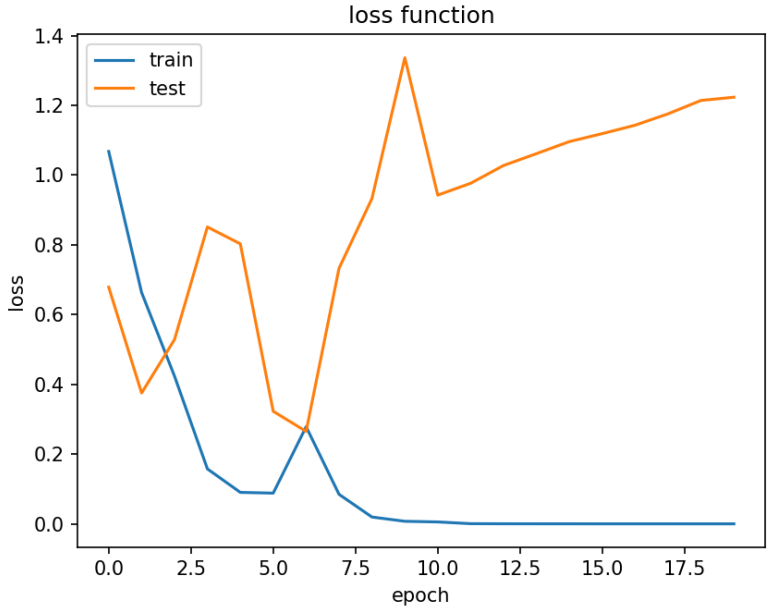
由于预测具有随机性，故展示三次运行结果

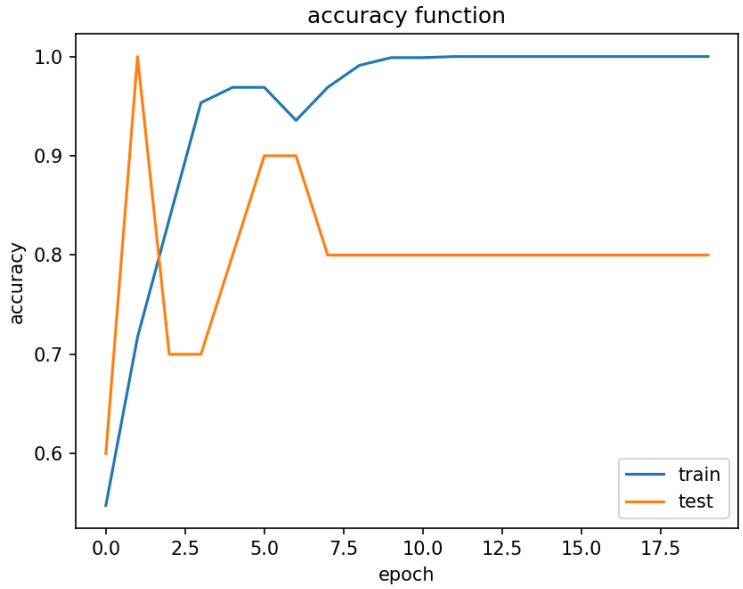
1



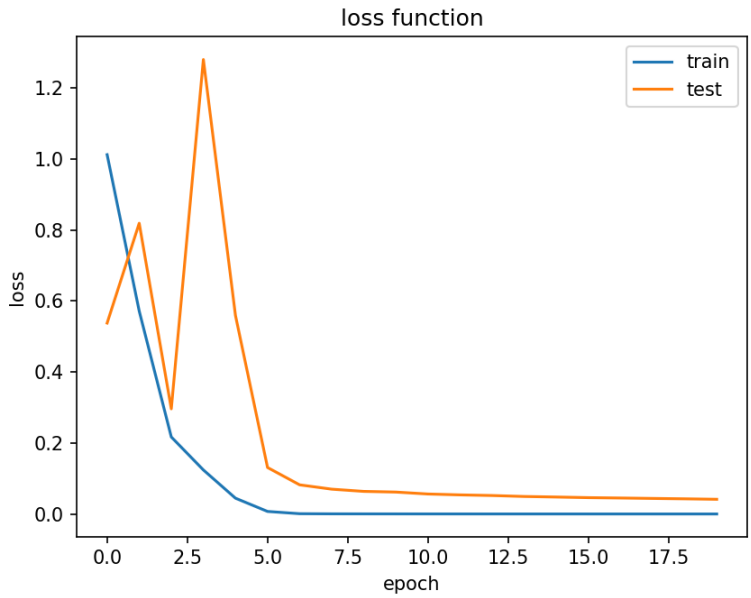


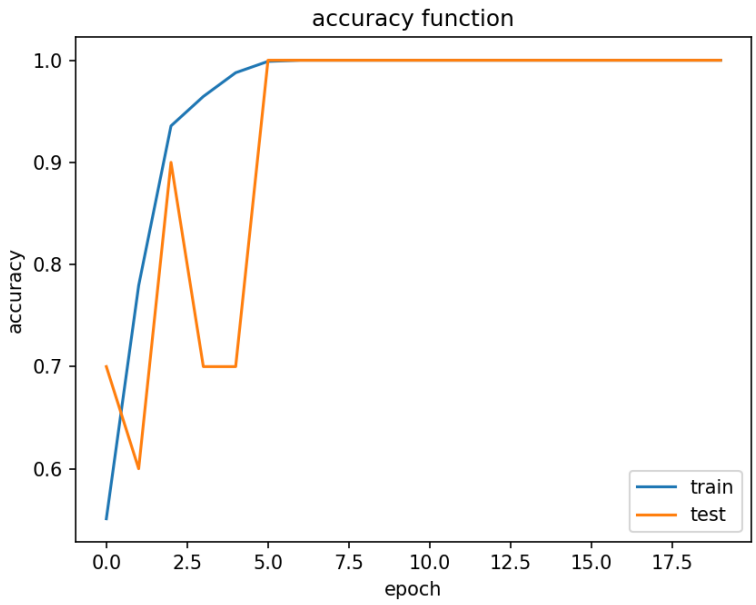
2


loss function
train
test
loss
epoch
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


accuracy function
train
test
accuracy
epoch
1.0
0.9
0.8
0.7
0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

3

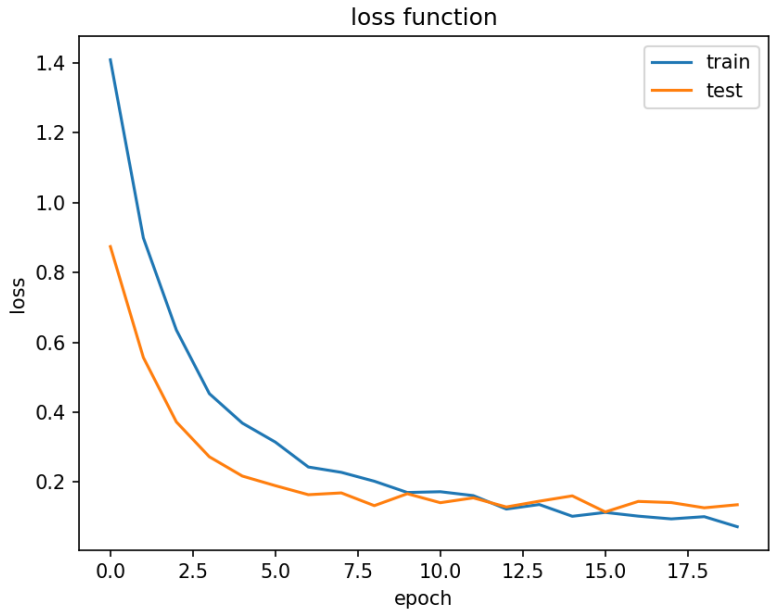


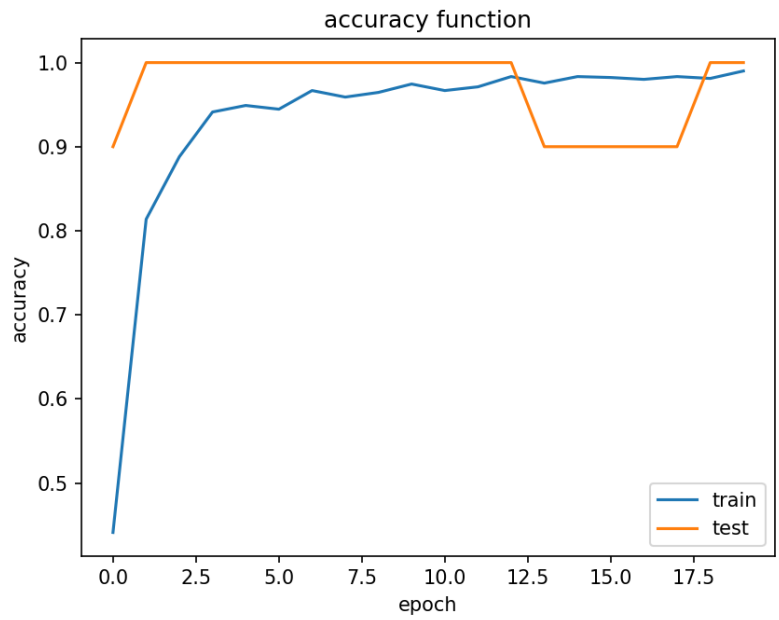


> `Resnet18`网络 + SGD

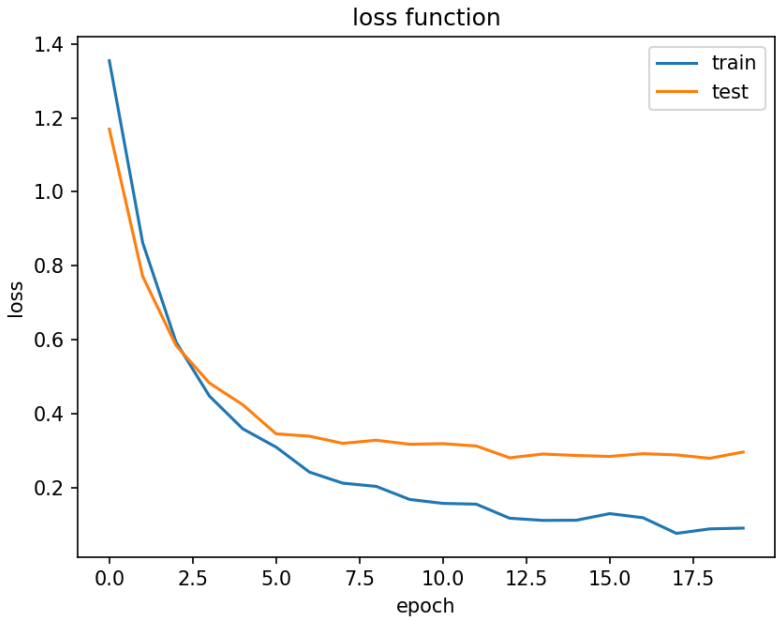
由于预测具有随机性，故展示三次运行结果

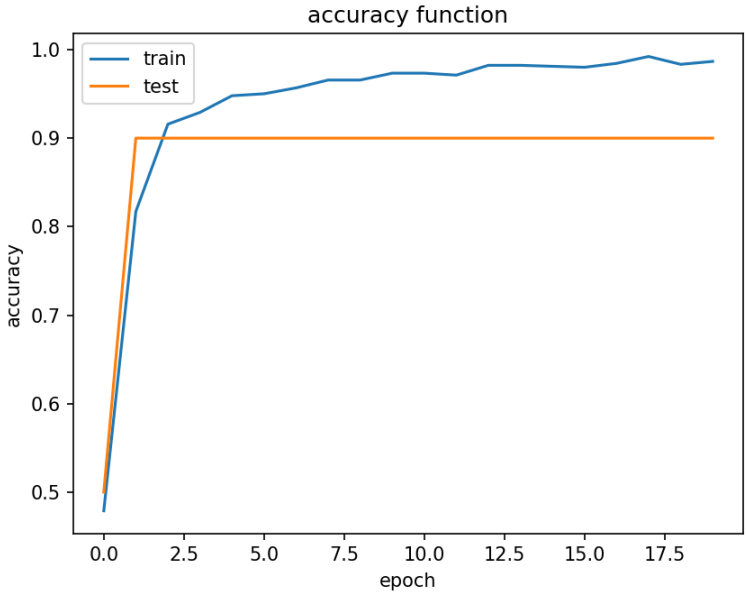
1


loss function
train
test
loss
epoch
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

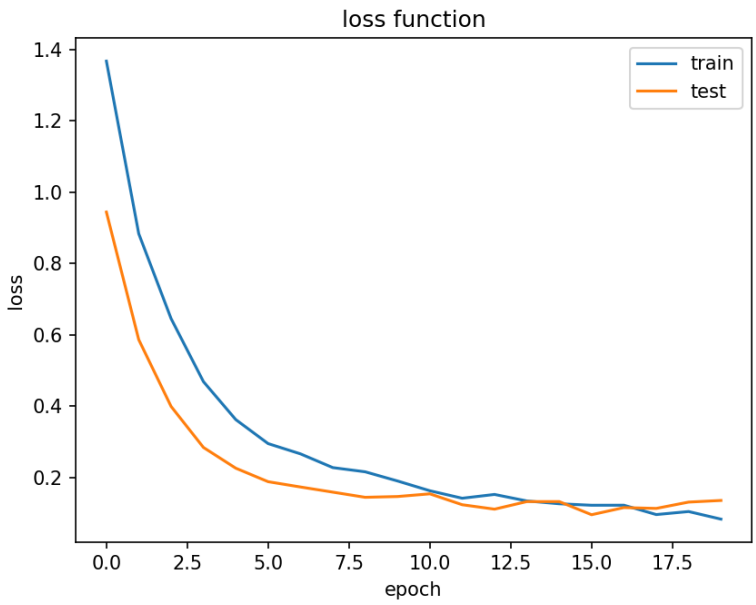

accuracy function
train
test
accuracy
epoch
1.0
0.9
0.8
0.7
0.6
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

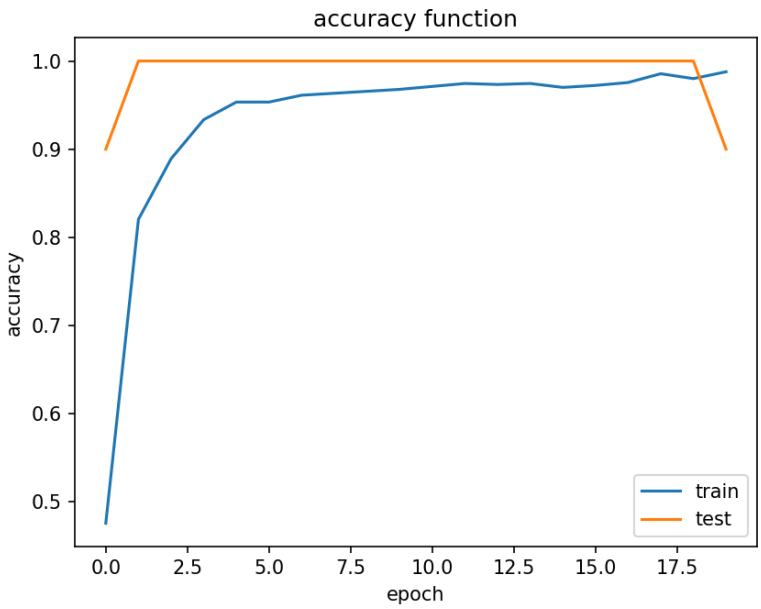
2


loss function
train
test
loss
epoch
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


accuracy function
train
test
accuracy
epoch
1.0
0.9
0.8
0.7
0.6
0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

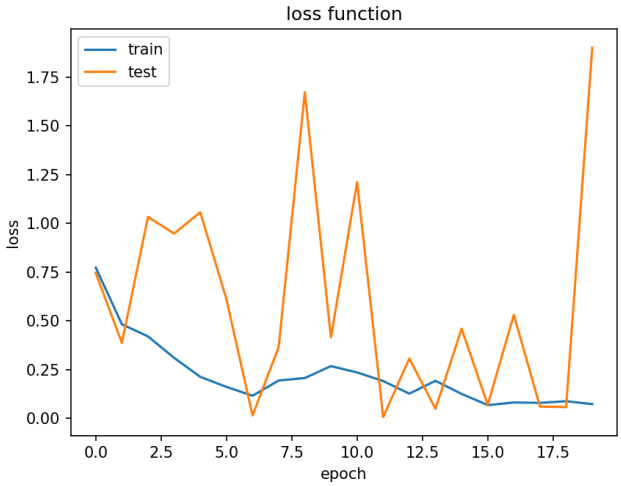
3

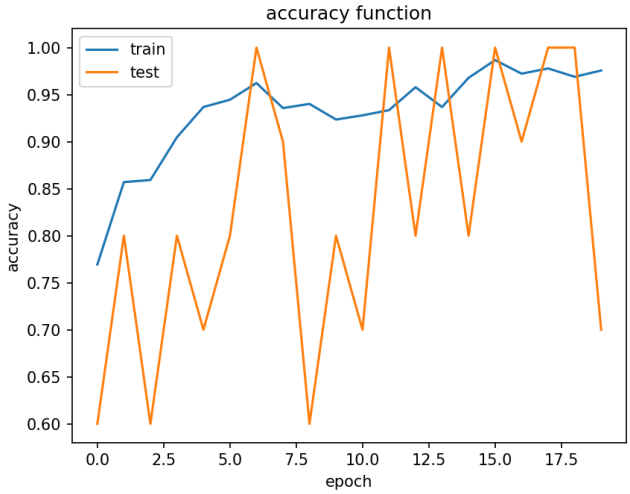




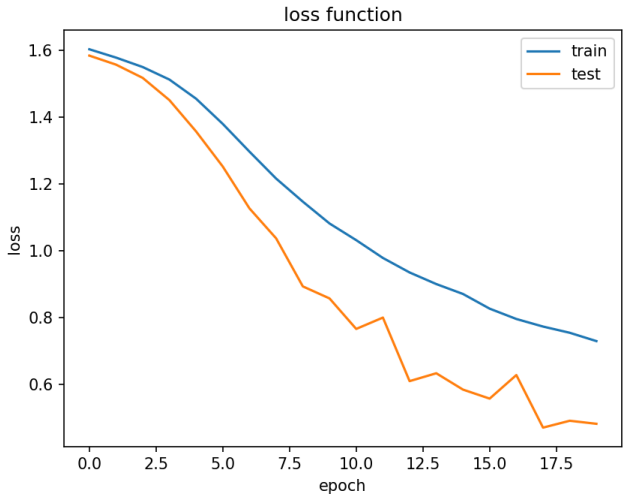
接下来说明如果混搭的后果：

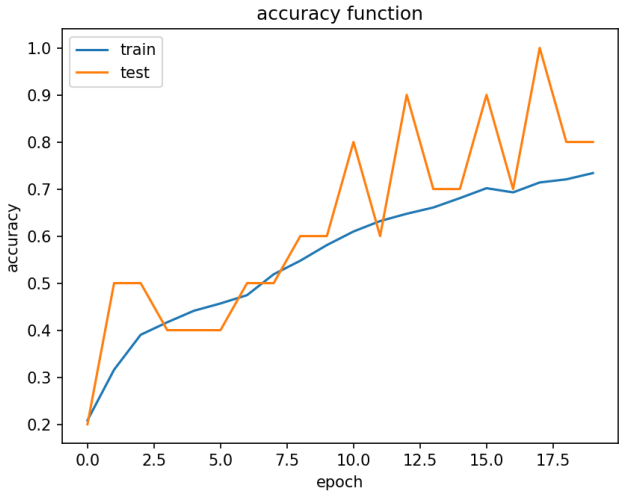
> `Resnet18` 网络+ Adam


loss function
train
test
loss
epoch
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


accuracy function
train
test
accuracy
epoch
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

卷积网络 + SGD





结合上述图像，不难发现测试集准确率波动很大且很频繁，使得结论不太稳定。

总体分析，(`Reanet18` + SGD) > (卷积 + Adam) > (卷积 + SGD) > (`Resnet18` + Adam)

但这只是基于上述层数设计的卷积网络，可能还具有很大的优化空间。如果继续优化网络结构，卷积的性能可能进一步提升。

四，思考题

此次实验没有思考题

五，参考资料

1. 2.CSDN